Question - 1

Converting operands to 8 bit binary

+25: 16 + 8 + 1

↳ 00011001

-25:

1's Complement of +25: 11100110

Add 1: 11100111

+18: 16 + 2

↳ 00010010

-18:

1's complement : 11101101

Add 1 : 11101110

(a) (25+(+18)) => 00011001
00010010
00101011

Binary result: 00101011

Decimal : 32+8+2+1
= 43

overflow check: Result is positive

No overflow

b) $((-25) + (-18))$

$$\begin{array}{r} 11100111 \\ +\ 11101110 \\ \hline (1)11010101 \end{array}$$

Binary result: 11010101

Decimal conversion: 00101010

Add 1: 00101011

MSB = 1

Value = -43

Overflow check: Result is negative

No overflow

Question- 2

(a) logical left shift of 10110110 by 2 positions

original: 10110110

shift by 1: 01101100

shift by 2: 11011000

final 8-bit result: 11011000

(b) logical right shift of 11010100 by 2 positions

original: 11010100

shift by 1: 01101010

shift by 2: 00110101

final 8-bit result: 00110101